



ABES Engineering College, Ghaziabad

Course Title: Basics of Mechanical and Automation Engineering

Course Code: 25ME101

Course Credits: 3

L T P

2 1 0

Minimum Required Hours (Theory): 40 Hrs

Pre-requisites: Basic knowledge of Physics and Mathematics

Course Level: Undergraduate

Course Objective:

The course aims to build interdisciplinary awareness and practical insight into modern engineering applications. It provides all engineering students with a strong foundation in key concepts of Mechanical Engineering and Automation, applicable across domains such as design, manufacturing, thermal, energy systems, automation, and control engineering.

Course Outcomes: After the completion of this course, students will be able to

Course Outcomes		K Level
CO1	Apply the principles of engineering mechanics and analyze mechanical systems using concepts of force, stress-strain and material properties.	K3
CO2	Analyze thermodynamic systems and apply the laws of thermodynamics to basic energy systems, including HVAC applications.	K3
CO3	Explain the construction, working principles and comparative aspects of IC engines, electric vehicles, and hybrid vehicles.	K2
CO4	Describe the working principles and functions of sensors, actuators, and control elements used in basic automation and mechanical actuation systems.	K2
CO5	Identify and interpret the applications of robotics and 3D printing in smart and automated industries.	K2

Content	
Unit I	Introduction to Mechanics Engineering Mechanics: Force moment and couple, principle of transmissibility, Resultant of force system- concurrent Types of supports (Hinge, Roller) and loads (Point, UDL, UVL), free body diagram, equilibrium equations and Support Reactions. Strength of Materials: Normal and shear Stress, Strain, Hookes' law, Poisson's ratio, elastic constants and their relationship, stress-strain diagram for ductile and brittle materials, factor of safety.
Unit II	Basic Thermodynamics Thermodynamics: Introduction to Thermodynamics, system and surrounding, basic thermodynamic terminologies (equilibrium, state, property, process, cycle), reversible and irreversible process, laws of thermodynamics (with numerical on first law of thermodynamics), comparison among heat engine and heat pump. Refrigeration: Refrigerating Effect, Ton of Refrigeration, Coefficient of Performance, Principle of Vapour Compression Refrigeration System, Construction, and Working of Domestic Refrigerator.

Air-Conditioning: Introduction and Applications of Air Conditioning, Psychrometric Properties and Construction and Working of Window Air Conditioner.	
Unit III	Internal Combustion Engine & Electric Vehicle
IC Engine: Definition and Basic Components of an Engine, Construction and Working of Two-Stroke and Four-Stroke Spark and Compression Ignition Engines, Comparison Between Two-Stroke and Four-Stroke SI and CI Engines, Types of fuels. Electric vehicles and hybrid electric vehicles: Working principle and components of an EV, advantages and disadvantages of EVs. working of hybrid electric vehicles, series and parallel types of HEV, degree of hybridization, advantages and disadvantages of HEV.	
Unit IV	Introduction to Automation and Control
Sensors and Transducers: Introduction to Sensors, transducers and their applications, basics of Feedback and Control: Open-loop vs Closed-loop Systems. Overview of Mechanical Actuation System: Overview of kinematic chains, cam mechanisms, ratchet mechanisms, gears and their types, belts, and bearings. Hydraulic and Pneumatic Actuation Systems: Overview of fluid power systems, Pressure and direction control valves, rotary actuators, basic hydraulic and pneumatic sequencing problems.	
Unit V	Emerging Trends in Engineering
Robotics: Introduction and significance in automation, Classification (articulated, cartesian, cylindrical robots, etc.); Robot specifications (DOF, payload capacity, reach, repeatability), Introduction to SCARA robots. 3D Printing: Introduction and types of 3D printing technologies, materials used (plastics, metals, composites), 3D printing process workflow, applications and limitations.	

Text Reading:

- 1) Engineering Mechanics by S.S. Bhavikati & K.G. Rajasekharappa, New Age International (P) Limited
- 2) Strength of Materials by R.K. Rajput, S. Chand
- 3) Engineering Thermodynamics by R. K. Rajput, Laxmi Publications
- 4) Refrigeration and Air Conditioning by R.S. Khurmi, S. Chand
- 5) Internal Combustion Engines by V. Ganesan, McGraw-Hill Education
- 6) Electric and Hybrid Vehicles by Amir Khajepour, Wiley Publications.
- 7) Industrial Automation and Robotics by Mikell P. Groover, Pearson Education
- 8) 3D Printing and Additive Manufacturing by Ian Gibson, David Rosen, Brent Stucker, Springer

Reference Reading:

1. Engineering Mechanics by S. S. Bhavikatti, New Age International Publications.
2. Strength of Materials by S.S. Rattan, McGraw-Hill
3. Refrigeration and Air Conditioning by C.P. Arora, McGraw-Hill
4. Electric and Hybrid Vehicles by Iqbal Husain / M. Ehsani, CRC Press / Springer
5. Fundamentals of Electric Vehicles Technology by K.T. Chau, CRC Press
6. Mechatronics Systems by Rolf Isenmann, Springer.
7. Introduction to Industrial Robotics, by Nagrajan, Pearson India.
8. Robotics and Industrial Automation by R. K. Rajput, S. Chand Publication